

LIVE DEMONSTRATION · JUNE 2026

The Universal Data Router

Ask anything. **Orbital routes the data.**

A working demonstration of Orbital as a company, personal, markets & code "brain" — assembling live answers across many independent APIs it was never told how to connect.

Prepared for Marty — Orbital · Sebastian Gray — Big Ideas Group

Built on github.com/jhammant/orbital-api-taxonomy · orbitalhq.com

THE PROBLEM

Every cross-system question is an integration project.

"Give me a full profile of this company — who they are, where they are, the risk & context around them." In most companies that's a two-quarter project: five teams, five API contracts, brittle glue, a new pipeline, and a data model nobody agrees on. Add a sixth source — start over.

OLD WAY • IPAAS / ETL

$O(N^2)$

A human pre-wires every $A \rightarrow B$ connection. N sources $\rightarrow N^2$ brittle connectors to build and maintain. Data is copied on a schedule and goes stale. New source = new pipeline.

THE TAX

"We can't answer that."

Integration sprawl makes the platform team the bottleneck every other team complains about — and every new question waits in their backlog.

ORBITAL • SEMANTIC ROUTER

$O(N)$

Describe each source once by the **meaning** of its data. Orbital derives the call graph at query time. The N th source is instantly reachable from every existing query.

Route data by **meaning**, not by plumbing.

- 01 **Shared semantic types.** Describe each API's inputs & outputs with a common vocabulary — Postcode, Latitude, CompanyName, Ticker.
- 02 **Describe, don't integrate.** A few lines of Taxi per source. No point-to-point code between any two services — ever.
- 03 **Query-time composition.** Orbital matches one operation's output type to another's input and plans the multi-API call graph itself.
- 04 **Lineage as proof.** Every assembled answer ships with a deterministic, auditable graph of exactly how it was built.

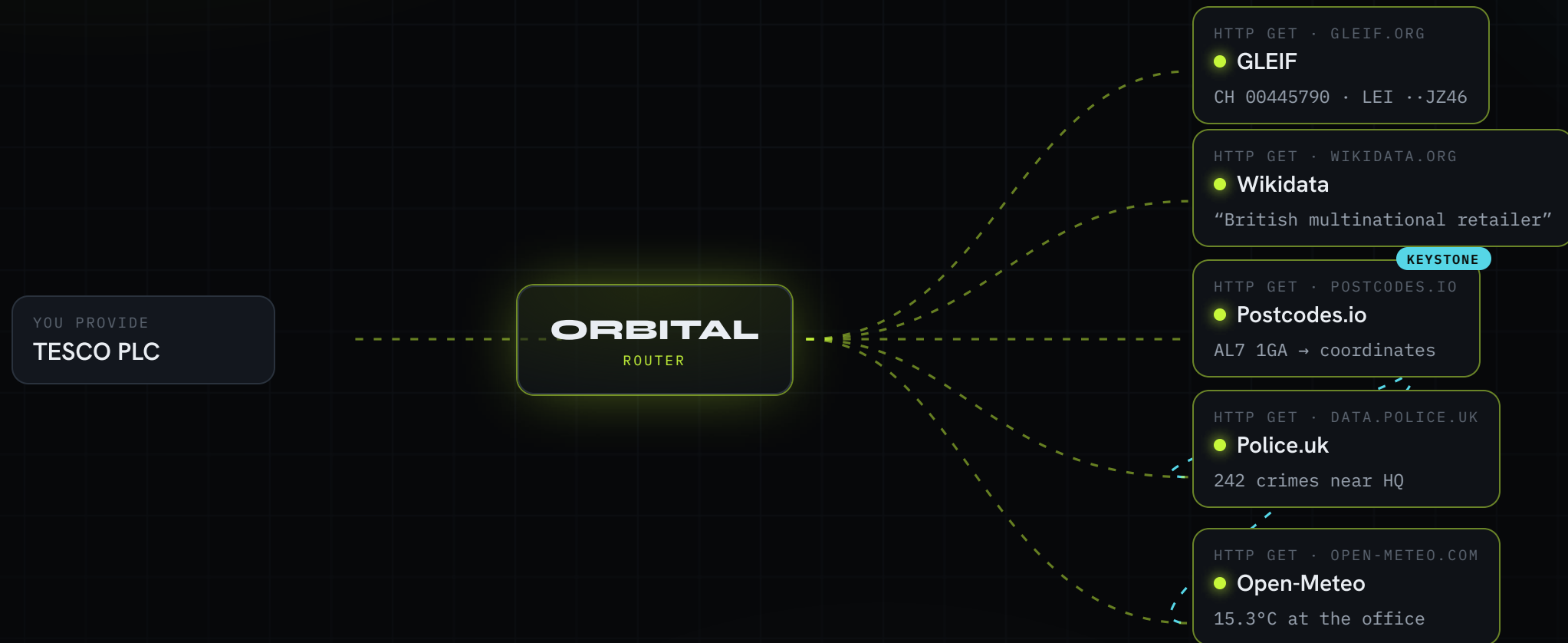
THE WHOLE IDEA

**One engine.
One spine.
Every
domain.**

A company links to its ticker, its filings, its people, its repos — all the same entity, answerable as one.

LIVE PROOF · CAPTURED FROM THE RUNNING STACK

Type a company name. 5 live APIs, 5 seconds, 0 integrations.



assembled in 5.0s · 5 live services routed · 0 integrations written · Orbital was never told these APIs connect – it deduced the path from shared types.

One tiny adapter fused two whole clusters.

The model had an **identity cluster** (company registries, emitting a postcode) and a **geo cluster** (crime, flood, weather, transit — all keyed on coordinates) that never touched. One ~30-line adapter — Postcodes.io, turning a postcode into coordinates — became the single edge that connected them.

IDENTITY CLUSTER

- Companies House
- GLEIF (LEI)
- Wikidata

emits → Postcode

Postcodes.io
THE KEYSTONE

Postcode → Latitude / Longitude

GEO CLUSTER

- Police.uk (crime)
- Environment Agency (flood)
- Open-Meteo (weather)

needs ← Latitude / Longitude

The Nth source you describe becomes reachable from queries involving all N-1 prior ones — and makes them richer — at **linear cost**. That compounding vocabulary is a network effect a pile of point-to-point integrations can never become.

SAME BRAIN, EVERY DOMAIN

One router. A market in every direction.

COMPANY / KYB

LIVE

TESCO PLC

Identity · background · crime · flood · weather across 5 APIs from a name. Lineage = a native audit trail.

GLEIF Companies House +3

MARKETS

LIVE

AAPL · 306.31

Ticker → live quote, filings & news. Resolves to the same legal entity as the company brain via the ticker → LEI spine.

Yahoo SEC EDGAR paper trades

CODING

LIVE

356★ · react 245k★

Repo → issues, PRs, CI & past decisions → a typed brief. The dev domain, same engine.

GitHub claude-history

PERSONAL

NEXT

Brief my 3pm

Calendar → attendees → their companies → news → a meeting brief. Local-first, your data never leaves.

Calendar Gmail Drive

Each vertical reuses the same spine and instantly composes with the others — **company** ✕ **markets** ✕ **code** ✕ **personal**. The TAM expands every time a new source is described, not rebuilt.

Not an iPaaS. A semantic router.

CATEGORY	WHAT THEY DO	WHY ORBITAL IS DIFFERENT
iPaaS · Zapier / Mulesoft	A human pre-builds every A→B flow.	Derives the path at query time from shared types — nothing is pre-wired.
ETL / warehouse	Copies data on a schedule; answers go stale.	Calls sources live, on demand. No copy, no staleness, no central store to govern.
LLM “agent calls tools”	A model guesses tools; opaque, non-deterministic.	Deterministic, type-checked plan — with a receipt for every hop.

Moat 1 – Compounding taxonomy

Whoever curates the canonical vocabulary for a vertical owns the substrate every future integration snaps into. Open schemas accelerate it; the routing engine stays the product.

Moat 2 – Lineage as proof

The only approach that emits deterministic, per-answer provenance. The "wow" of the demo, the technical buyer's trust, and the compliance buyer's deliverable — one feature.

LEAD · COMPLIANCE / KYB

Lineage = the audit trail

Customer/supplier 360 with native provenance regulators ask for.

CHAMPION · DATA PLATFORM

$O(N^2) \rightarrow O(N)$

Connector backlog collapses to type descriptions; they curate, not glue.

VOLUME · REVOPS

Live account enrichment

Firmographics + territory + signals as a one-line query, always fresh.

Built live, in one working session.

- ✓ **The keystone bridge.** Postcodes.io adapter that fuses the identity & geo clusters — verified live.
- ✓ **No-auth "type a company name."** GLEIF resolves name → LEI → registry number → postcode, no keys.
- ✓ **6 new live adapters.** GLEIF · Wikidata · Open-Meteo · Yahoo · GitHub · GDELT — on top of 9 UK-gov APIs.
- ✓ **A polished demo app.** "The Company Brain" — live routing animation, dossier, trading & coding tabs.
- ✓ **Visible lineage.** Orbital's own Profiler graph shows the multi-API route, captured live.

WHAT INVESTMENT ACCELERATES

PUBLIC SANDBOX	"type any company", hosted
NEW VERTICALS	Finance · Health (FHIR)
BRAIN-SMARTS	ask-in-English · watch & alert
ENTERPRISE	governance · PII policies
NETWORK EFFECT	the Data Router Challenge

The wedge: developer-loved open semantic schemas; the routing engine is the product; the taxonomy is the moat that compounds with every contribution.

Ask anything. It routes the data.

The integration era was about connecting apps you tell it to connect. Orbital connects apps it **figures out** how to connect — by meaning — and proves the path. Every source you teach the brain makes every question smarter, at linear cost, with a receipt.

Live demo `localhost:8848` · the Company Brain

Engine `orbitalhq.com` · Schemas `github.com/jhammant/orbital-api-taxonomy`